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Studierichting: Informatica
Oefeningenreeks 2D nr. 10.13

$$\lim_{x \to \frac{1}{4}} \frac{4x^3 + 13x^2 + \frac{5}{2}x - \frac{3}{2}}{3x^2 + \frac{13}{4}x - 1}$$

$$= \frac{0}{0}$$

Teller ontbinden in factoren

$$\begin{array}{c|cccc}
 & 4 & 13 & \frac{5}{2} & \frac{-3}{2} \\
\frac{1}{4} & & 1 & \frac{14}{4} & \frac{3}{2} \\
\hline
 & 4 & 14 & \frac{24}{4} & \multicolumn{1}{|c}{0}
\end{array}$$

Teller $= (4x-1)\left(x^2 + \frac{14}{4} + \frac{3}{2}\right)$

Noemer ontbinden in factoren

$$\begin{array}{c|ccc}
 & 3 & \frac{13}{4} & -1 \\
\frac{1}{4} & & \frac{3}{4} & 1 \\
\hline
 & 3 & 4 & \multicolumn{1}{|c}{0}
\end{array}$$

Noemer $= (4x-1)\left(\frac{3}{4}x + 1\right)$

$$\lim_{x \to \frac{1}{4}} \frac{(4x-1)\left(x^2 + \frac{14}{4} + \frac{3}{2}\right)}{(4x-1)\left(\frac{3}{4}x + 1\right)}$$

$$= \frac{39}{19}$$

# References